



Yarmouk University

Hijjawi Faculty for Engineering Technology

Computer Engineering Department

CPE 563: Computer Networking Laboratory

Task 10: Ethernet & ARP

Basic Information:

Student Name:	
Student ID:	
Student Section Number:	
PC Logical Address:	
PC Physical Address:	

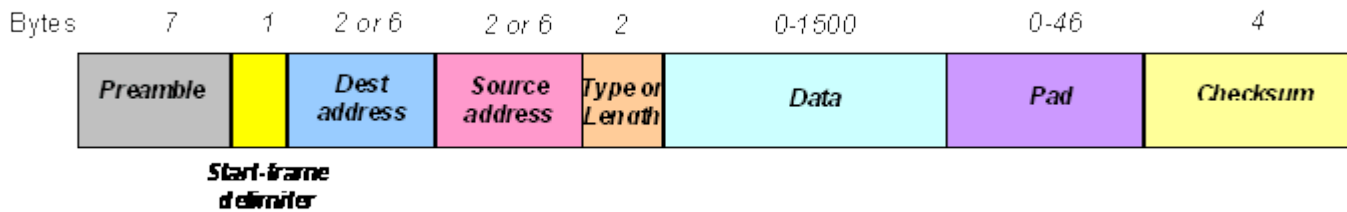
Objective:

In this lab, we'll explore two important protocols which are Ethernet and ARP:

1. Learn the student how Layer 2 works using Ethernet protocol.
2. Learn about ARP protocol.

Background:

Ethernet Frame



- The Preamble consists of seven bytes all of the form 10101010, and is used by the receiver to allow it to establish bit synchronization (there is no clocking information on the Ether when nothing is being sent).
- The Start frame delimiter is a single byte, 10101011, which is a frame flag, indicating the start of a frame.
- The MAC addresses used in 802.3 are always 48 bits long, although older versions of Ethernet used 16 bits. By convention, Ethernet addresses are usually quoted as a sequence of 6 bytes (in hexadecimal) with each byte quoted with its bits in reverse order (this curious arrangement is driven by the transmission order). Individual addresses have a most significant bit of 0, multicast addresses a most significant bit of 1 (the most significant byte is thus quoted as xxxx xxx1). An address of 48 0s is a broadcast to all stations on the local network. An interesting feature is that individual addresses may be local or global, with, respectively, a second most significant bit of 0 or 1 (so a most significant byte of xxxx xx0x is global). Local addresses have no significance except on the local Ethernet installation, but global addresses are unique: every system with an Ethernet interface has a unique global address hardwired into that interface. In principle, any station can address any other, anywhere in the world, but of course this would rely on a rather impractical internet layer to identify the target, and perform the routing. On a single Ethernet there is no such problem because a frame is seen by all stations and each can recognise its own address. Note that these are addresses of SAPs at the top of the MAC sublayer and it is to these SAPs that Ethernet delivers frames: from the point of view of the higher layers, these are the NPAs.
- The Length/EtherType field is the only one which differs between 802.3 and Ethernet II. In 802.3 it indicates the number of bytes of data in the frame payload, and can be anything from 0 to 1500 bytes. Frames must be at least 64 bytes long, not including the preamble, so, if the data field is shorter than 46 bytes, it must be compensated by the Pad field. The reason for specifying a minimum length lies with the collision-detect mechanism. In CSMA/CD a station must never be allowed to believe it has transmitted a frame successfully if that frame has, in fact, experienced a collision. In the worst case it takes twice the maximum propagation delay across the network before a station can be sure that a transmission has been successful. If a station sends a really short frame, it may actually finish sending and release the Ether without realising that a collision has occurred. The 802.3 design rules specify an upper limit on the maximum propagation delay in any Ethernet installation, and the minimum frame size is set to be more than twice this figure (64 bytes takes 51.2ms to send at 10Mbps).

- In Ethernet II, on the other hand, this field is used to indicate the type of payload carried by the frame. For example 080016 signifies an IP payload. In fact the smallest legal value of this field is 060016, and since the greatest value of the 802.3 Length is 05DC16 it is always possible to tell Ethernet and 802.3 frames apart and they can therefore coexist on the same network. 802.3 was intended to be used with 802.2 LLC as its standard payload, the latter using a 7-bit subaddress to specify protocol type. However, this is not compatible with the 16-bit EtherType of Ethernet II, so the SNAP (Subnetwork Access Protocol) extension was developed. With a SNAP-extended header, an LLC PDU can carry a 16-bit EtherType.
- Finally the Checksum field uses a CRC-32 polynomial code.

Lab Work:

Capturing and analyzing Ethernet frame

- Clear your browser cache.
- Enter the following URL into your browser
<http://gaia.cs.umass.edu/wireshark-labs/HTTP-ethereal-lab-file3.html>
 your browser should display the rather lengthy US Bill of Rights.
- Stop Wireshark packet capture.
- Find HTTP GET message and the HTTP OK message and record the packet numbers on a sheet for your reference.
- Since this lab is about Ethernet and ARP, we're not interested in IP or higher layer protocols. So we'll change Wireshark's "listing of captured packets" window so that it shows information only about protocols below IP. To do this, select Analyze->Enabled Protocols. Then uncheck the IPv4 box and select OK.
- Select the Ethernet frame containing the HTTP GET message. You can use the packet number you recorded before or look into the packet contents windows (lower display windows in Wireshark) and search for a frame with GET method in its data.

Answer the following questions based on the contents of the Ethernet frame containing the HTTP GET message:

1. What is the 48-bit Ethernet address of your computer?
2. What is the 48-bit destination address in the Ethernet frame? Is this the Ethernet address of gaia.cs.umass.edu? (Hint: the answer is no). What device has this as its Ethernet address?
3. Give the hexadecimal value for the Frame type field. What upper layer protocol does this value corresponds to?
4. How many bytes from the very start of the Ethernet frame does the ASCII "G" in "GET" appear in the Ethernet frame?

Answer the following questions based on the contents of the Ethernet frame containing the HTTP response (OK) message:

5. What is the value of the Ethernet source address? Is this the address of your computer, or of gaia.cs.umass.edu (Hint: the answer is no). What device has this as its Ethernet address?
6. What is the destination address in the Ethernet frame? Is this the Ethernet address of your computer?
7. How many bytes from the very start of the Ethernet frame does the ASCII "O" in "OK" appear in the Ethernet frame?

Capturing and analyzing an ARP traffic

- Use the same trace of the previous part. Enter arp in the filter field.
- Find any ARP request message and its corresponding response message.

Answer the following questions:

8. What are the hexadecimal values for the source and destination addresses in the Ethernet frame containing the ARP request message?
9. Give the hexadecimal value for the two-byte Ethernet Frame type field. What upper layer protocol does this value corresponds to?
10. How many bytes from the very beginning of the Ethernet frame does the ARP opcode field begin?
11. What is the value of the opcode field?
12. Does the ARP request message contain the IP address of the sender?
13. Where in the ARP request does the “question” appear – the Ethernet address of the machine whose corresponding IP address is being queried?
14. For the ARP reply message what is the value of the opcode field?
15. Where in the ARP reply message does the “answer” to the earlier ARP request appear – the IP address of the machine having the Ethernet address whose corresponding IP address is being queried?
16. What are the hexadecimal values for the source and destination addresses in the Ethernet frame containing the ARP reply message?